

Experiment No. 17

Aim: To determine melting point of given organic compound.

Apparatus: Thiele's tube, retort stand with clamps thermometer, capillary tube, thread, burner etc.

Chemicals: Given organic compound, liquid paraffin

Theory: A pure organic compound possesses characteristic physical properties such as refractive index, specific gravity, melting point, boiling point etc. However for practical purpose in laboratory melting point and boiling point are used as criteria for purity. A pure organic solid has a definite and sharp melting point, while an impure substance has a lower and indefinite melting point. The melting point of a substance is the temperature at which the solid and liquid phases exist in equilibrium with each other.

Procedure: Take a small capillary tube and seal it at one end by heating. Fill the finely powdered dry organic compound (about 1 cm in height) in the capillary tube by tapping it on porcelain tile. Tie the capillary tube to the thermometer such that bulb of thermometer and organic substance are at the same level. Now, suspend the thermometer in Thiele's tube containing liquid paraffin such that it should not touch the sides of the Thiele's tube. Heat gently the side arm only with wavy flame. Stop heating Thiele's tube when solid starts melting. Record the temperature when solid is completely melts. This temperature is melting point of given organic compound.

Observation: Melting point of given compound = $\underline{122}^{\circ}\text{C}$
 $= (\underline{122}^{\circ}\text{C} + 273) \text{ K} = \underline{395} \text{ K}$

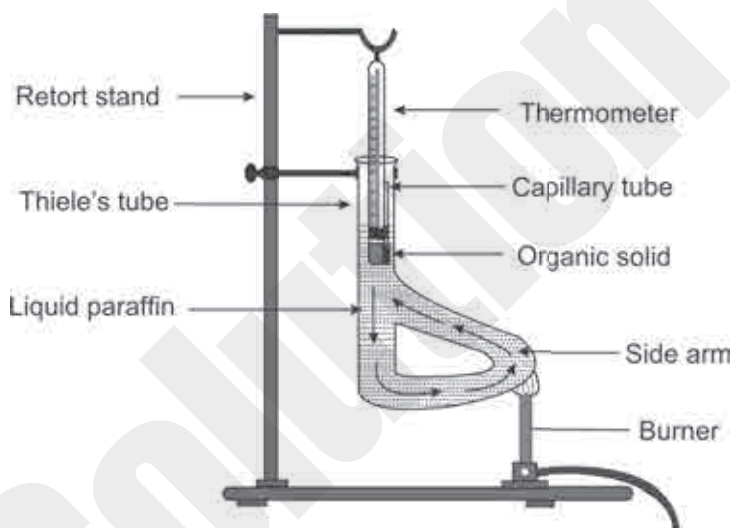


Fig. 17.1 Determination of melting point

Table: Melting point of some organic compounds

Sr. No.	Compounds	Melting Point ($^{\circ}\text{C}$)
1	Naphthalene	80
2	Oxalic acid	101
3	β -naphthol	122
4	Benzoic acid	122

Remark and sign of teacher:

.....

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. Melting point of a solid is

- a. The temperature at which a solid is in equilibrium with its liquid, at 1atm pressure.
- b. The temperature at which a liquid is in equilibrium with its gas, at 1atm pressure.
- c. The temperature at which the vapor pressure of the liquid is equal to atmospheric pressure.
- d. The temperature at which solid in capillary tube start melting

2. In the determination of melting point of a solid, the solid in small quantity is filled in a

- a. fusion tube
- b. capillary tube with one sealed end.
- c. Thiele's tube containing paraffin liquid.
- d. capillary tube with both open end.

3. Melting point is a criteria of purify of a organic solid because

- a. a pure liquid boil sharply at a definite temperature.
- b. a pure solid melts sharply at a definite temperature.
- c. a pure solid melts at some higher temperature than its melting point.
- d. a pure liquid boils at some lower temperature than its boiling point

4. The melting point of oxalic acid is

- a. 80 °C
- b. 101 °C
- c. 110 °C
- d. 122 °C

5. In the determination of melting point, Thiele's tube is filled with paraffin oil because

- a. paraffin oil has high boiling point
- b. paraffin oil is chemically highly reactive
- c. paraffin oil undergo decomposition
- d. paraffin oil react solid quickly.

6. The temperature in °C is converted into Kelvin by addition of

- a. 0
- b. 100
- c. 273
- d. 373

Short answer questions

1. Define melting point.

Ans.....

The melting point of a solid is the temperature at which the solid and its liquid are in equilibrium with each other at atmospheric pressure.

2. What is criteria of purity of solid substances?

Ans.....

A pure solid has a sharp and definite melting point. Therefore, the melting point is used as a criterion of purity of solid substances. Impurities lower and broaden the melting point range.

Remark and sign of teacher: